Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

ANS:

Windows IP Configuration

Host Name . . . . : Office-PC
Primary Dns Suffix . . . . :
Node Type . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : localdomain
Description . . . . : Intel(R) Wi-Fi 6 AX201
Physical Address. . . . . : A1-B2-C3-D4-E5-F6
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.1.45(Preferred)
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . : 192.168.1.1
DHCP Server . . . . : 192.168.1.1
DNS Servers . . . . : 8.8.8.8
8.8.4.4

Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

ANS:

Here is a simulation and explanation of what happens when you run a ping test with and without an active network connection:

1. Disconnected State (`ping [www.google.com](https://www.google.com)` without Wi-Fi/Ethernet)

- **What happens:** The command fails immediately or times out. You will typically see error messages such as:
    - `General failure.`

    - `Ping request could not find host [www.google.com](https://www.google.com). Please check the name and try again.` (indicating a DNS resolution failure because there is no path to a DNS server)
    - `Request timed out.` or `Destination host unreachable.`
- **What it means:** Your computer has no active physical or wireless link to a local network interface or gateway. Without a connection to the local router, it cannot route packets outward to the Internet or reach a DNS server to translate web addresses into IP addresses.

2. Connected State (`ping [www.google.com](https://www.google.com)` with active Wi-Fi/Ethernet)

- **What happens:** The command successfully sends Internet Control Message Protocol (ICMP) echo request packets to Google's servers and receives replies, displaying round-trip times (in milliseconds) and statistics (packets sent, received, and lost).
- **What it means:** Your local adapter is successfully linked to your gateway (router), which translates and forwards your traffic across the public WAN to the destination, confirming full end-to-end network and internet connectivity.

Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

ANS:

Here is how you can check it yourself and what typical values look like:

How to Check It on Your PC

1. **Open Task Manager:** Press `Ctrl + Shift + Esc`, then click on the **Performance** tab and select **Ethernet** or **Wi-Fi**, OR go directly to the **Processes** tab and click the **Network** column header to sort apps by bandwidth usage.
2. **Open Resource Monitor:** Alternatively, press `Win + R`, type `resmon`, and press Enter. Go to the **Network** tab to see real-time bandwidth consumption per process.
3. **Stream a Video:** Play a 1080p or 4K YouTube video in your browser for a few minutes and observe the spikes.

Typical Simulated Results (1080p YouTube Stream)

- **Process Using the Most Bandwidth:** Your web browser (e.g., `chrome.exe`, `msedge.exe`, or `firefox.exe`) streaming the video.
- **Peak Network Usage Value:**
    - For **1080p video:** Typically fluctuates between **3 Mbps to 8 Mbps** (kilobits/megabits per second), depending on buffering and codec efficiency (like AV1 or VP9).
    - For **4K video:** Can easily peak between **15 Mbps to 25 Mbps**.

Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.

ANS:

How to Filter and What You Will See

1. **Apply the Filter:** In Wireshark's top filter bar, type `tls` (or `http` if testing an unencrypted site) and press Enter.
2. **Three Types of Information Visible in the Filtered Packets:**
    - **IP Addresses and Port Numbers:** You will clearly see your computer's local IPv4 address as the source and the destination IP address of Flipkart's or Instagram's cloud/CDN servers, along with port **443** (the standard port for secure HTTPS traffic).
    - **Server Name Indication (SNI):** If you expand the **Client Hello** packet during the initial TLS handshake, you can see the cleartext domain name (such as `[www.instagram.com](https://www.instagram.com)` or `[www.flipkart.com](https://www.flipkart.com)`) before the encrypted session is established.
    - **Frame Length and Timestamps:** Every captured packet shows its exact size in bytes (e.g., length 517) and precise timestamps down to microseconds, letting you measure the exact duration of handshakes and data transfers.

Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.
Topic Repository URL

ANS:

Home Network Optimization Checklist for Online Gaming

1. **Switch to a Wired Ethernet Connection:** Bypass Wi-Fi completely by plugging an Ethernet cable directly from your PC or console into the router to eliminate wireless interference and reduce packet loss.
2. **Enable Quality of Service (QoS):** Log into your router's admin dashboard and activate QoS settings to prioritize gaming traffic over other household bandwidth usage (like streaming or large downloads).
3. **Power Cycle Your Hardware:** Unplug both your modem and router for about 60 seconds, then plug them back in to clear temporary memory leaks and refresh your connection with the ISP.
4. **Change Your DNS Servers:** Switch to faster, more reliable public DNS servers like Cloudflare (`1.1.1.1`) or Google (`8.8.8.8`) to speed up initial domain and game server lookups.
5. **Close Background Bandwidth Hogs:** Shut down active background applications, torrent clients, cloud sync tools, or browser tabs streaming video on your local machine.